

CLASS 9 · CBSE · GANITA MANJARI PART 2 · DPP SHEET

Chapter 12: Quadrilaterals

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In a parallelogram $ABCD$, $\angle A = 65^\circ$. Find the other three angles.
2. Two opposite sides of a parallelogram are $(4x + 3)$ cm and $(6x - 7)$ cm. Find x and the length of the side.
3. The diagonals of a parallelogram $ABCD$ meet at E . If $AC = 18$ cm and $BD = 12$ cm, find AE and DE .
4. The angles of a quadrilateral, in order, are 70° , 110° , 70° and 110° . Is it a parallelogram?
5. The sides of a triangle are 8 cm, 12 cm and 16 cm. Find the perimeter of the triangle formed by joining the mid-points of its sides.
6. D and E are the mid-points of AB and AC in $\triangle ABC$, and $BC = 15$ cm. Find DE .
7. A median of a triangle is 15 cm long. Into what lengths does the centroid divide it?
8. The diagonals of a quadrilateral are 10 cm and 24 cm. Find the perimeter of its Varignon parallelogram.
9. The diagonals of a quadrilateral are equal. What special kind of parallelogram is its Varignon parallelogram?
10. Can a regular octagon tile the plane on its own? Each of its angles is 135° .

Answer Key

1. $\angle B = 115^\circ, \angle C = 65^\circ, \angle D = 115^\circ$
2. $x = 5; 23 \text{ cm}$
3. 9 cm and 6 cm
4. Yes. Both pairs of opposite angles are equal.
5. $4 + 6 + 8 = 18 \text{ cm}$
6. 7.5 cm (and $DE \parallel BC$)
7. 10 cm (from the vertex) and 5 cm
8. $10 + 24 = 34 \text{ cm}$
9. A rhombus
10. No. 360 is not a multiple of 135.